

IPSB / RDQM Monitoring Modernization Initiative

Executive Overview

This initiative proposes a next generation monitoring architecture for IPSB environments running IBM MQ RDQM clusters. The objective is to replace hundreds of fragmented monitoring checks with a unified, intelligent health state system that is more scalable, easier to maintain, and faster to troubleshoot. The new design focuses on reliability, clarity of alerting, intelligent routing, and operational efficiency while minimizing monitoring noise and preventing unnecessary escalations.

Strategic Objectives

- 1 Reduce monitoring complexity across all sites
- 2 Improve accuracy of alerts and reduce false positives
- 3 Route incidents directly to teams capable of remediation
- 4 Provide operators with actionable diagnostics and recovery guidance
- 5 Ensure zero disruption when new applications or queue managers are introduced
- 6 Standardize monitoring logic across all environments
- 7 Support future growth without redesign

Current Operational Challenges

- 1 Hundreds of individual monitors create administrative overhead
- 2 Alert routing frequently requires manual triage
- 3 Monitoring logic differs between sites
- 4 Adding new applications requires manual monitoring configuration
- 5 Limited built in diagnostics when failures occur
- 6 No centralized model for health state assessment

Proposed Monitoring Architecture

The new framework introduces a single monitoring engine executed every 10 minutes under the mqm user context. Rather than monitoring individual metrics separately, the system evaluates environment health across structured domains and produces consolidated results for Zabbix inspection.

Health Domains

- 1 RDQM Cluster Health
- 2 Queue Manager Runtime Health
- 3 Queue Flow Health
- 4 Channel Health

- 5 Dependency Health (e.g., Connect:Direct)
- 6 Certificate/TLS Health

Each domain produces a status value and diagnostic details. A unified roll up state determines operational severity.

Dynamic Discovery Model

The system automatically discovers queue managers present on a server and determines which template rules apply based on naming conventions. Unknown applications automatically inherit a safe default monitoring template. This ensures new applications can be deployed without requiring monitoring configuration updates.

Alerting and Priority Model

The monitoring system aligns with ServiceNow priority classifications and organizational incident response policy.

- 1 P2 – Critical operational issue requiring prompt attention
- 2 P3 – Risk condition or upcoming problem
- 3 P4 – Informational or awareness notification
- 4 P1 – Reserved for manual command center escalation

Alerts default to the Operations team unless explicitly configured otherwise. Routing can be defined per monitored item, allowing specific conditions to be sent directly to application teams when appropriate.

Intelligent Routing Capability

Routing decisions are rule based and configurable. For example, a queue depth issue can be routed to an application team, while a channel failure in the same queue manager may still route to Operations. This allows precise engagement of responsible teams while avoiding unnecessary escalations.

Self Healing Policy

Automated corrective actions are supported but disabled by default. They must be explicitly enabled and are subject to rate limits and safety thresholds. This ensures that automation enhances stability without risking unintended service impact.

Integrated Local Knowledge Base

The solution includes a built in diagnostic guidance capability. Operators can run a command to receive detailed troubleshooting instructions tailored to the detected issue, including probable causes, verification steps, and recommended actions. This reduces mean time to resolution and ensures consistent troubleshooting practices across teams and sites.

Configuration Strategy

The system uses a layered configuration model:

- 1 Global configuration defines standard behavior and templates
- 2 Site configuration automatically generated on first run
- 3 Local overrides allow environment specific tuning
- 4 Rules merge predictably so updates never overwrite local customizations

Scalability and Future Readiness

- 1 Supports unlimited sites without redesign
- 2 Automatically adapts to new queue managers
- 3 Allows new monitoring logic without rewriting framework
- 4 Maintains stable interface for monitoring tools
- 5 Designed for long term maintainability

Operational Benefits

- 1 Fewer monitors to maintain
- 2 Faster incident detection
- 3 Faster root cause identification
- 4 Reduced alert fatigue
- 5 Improved team efficiency
- 6 Standardized monitoring across enterprise
- 7 Lower operational risk

Implementation Approach

Development will proceed in controlled phases beginning with a core monitoring framework in a test environment. Additional capabilities will be added incrementally to ensure stability and validation at each step. All scripts will execute under the mqm user account to match operational troubleshooting context.

Conclusion

This initiative establishes a modern monitoring architecture designed for reliability, clarity, scalability, and operational effectiveness. It replaces fragmented monitoring with a structured health state system that enables teams to detect issues faster, respond more efficiently, and scale monitoring seamlessly as environments evolve.